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Detection of LLM Hallucinations Using Late Internal Representations

Sakhawat Hossan Jing Deng
Department of Computer Science, UNC Greensboro, Greensboro, NC, U. S. A.
s_hossan@uncg.edu, jing.deng@uncg.edu

***Abstract*—Despite the impressive capabilities of Large Language Models (LLMs), their tendency to generate hallucinations, i.e., factually incorrect or nonsensical content, remains a major challenge for reliable deployment, especially in high-stakes fields such as healthcare, law, and scientific research. Detecting and reducing these hallucinations is critical for building more trustworthy systems. In this work, we propose a hallucination detection framework using a custom-designed classifier architecture that leverages late internal representations from LLM. We explore three distinct approaches for classifier input: (i) features from the Multi-Layer Perceptron (MLP) blocks, (ii) features from the Attention blocks, and (iii) a combination of both. Experiments across eight different LLMs and different evaluator scores show that our classifier, termed LIR, consistently outperforms other state-of-the-art approaches in AUC, correlation with human annotations, and response generation time. Our results demonstrate the effectiveness of using late internal representations for hallucination detection and contribute to the development of standardized benchmarks to evaluate the reliability of generative models.**

***Index Terms*—Large Language Models, LLM, MLP, Attention, Hallucination Classifier, NLP**

I. Introduction

LLMs have seen explosive growth and are now integrated into various aspects of daily life. From virtual assistants like ChatGPT and Google Bard to enterprise tools embedded in productivity suites, LLMs are reshaping how we access and generate information. Their ability to produce human-like responses has made them indispensable in areas such as content creation, tutoring, customer service, and decision support systems.

As this integration deepens, so do the stakes. In high-risk domains such as healthcare, legal services, and scientific research, the cost of incorrect or nonsensical responses is severe. A misdiagnosis from a clinical support chatbot, a flawed legal interpretation, or an incorrect citation in a literature review generated by an LLM can cause significant consequences. The growing reliance on generative systems has thus amplified concerns about factual reliability.

One major source of unreliability in LLM outputs is what researchers refer to as *hallucination* — a term describing the phenomenon where models generate fluent but factually incorrect or unsupported content. Although early neural machine translation models exhibited such behavior, the term gained popularity through the work of Lastras et al. [1], who formally analyzed hallucinations in neural machine translation. Their work prompted broader attention to the issue across natural language generation (NLG) domains.

Subsequent studies began to classify hallucinations and establish frameworks for detecting and mitigating them. Ji et al. [2] provided a comprehensive taxonomy of hallucinations, distinguishing between extractive and abstractive types, and between intrinsic and extrinsic causes. Rawte et al. [3] extended this effort, surveying hallucination behavior in large foundation models and highlighting limitations in existing evaluation benchmarks. These works laid the groundwork for understanding hallucination not just as a random error, but as a systemic property of generative models.

Early hallucination detection methods largely relied on *post-hoc* verification with external knowledge sources, such as retrieval-augmented generation [3]. While effective in some cases, these approaches introduced latency, scalability challenges, and a dependency on the completeness of external databases. This motivated a shift toward *internal-state* based detection, which leverages model activations to capture truthfulness signals embedded within the network. The INSIDE framework [4] demonstrated that activations from final decoder layers retain strong hallucination-related signals, while Orgad et al. [5] showed that intermediate layers encode truthfulness awareness even when it is absent in generated text, revealing a disconnect between model knowledge and output.

Several frameworks have since explored this direction. The MIND framework by Su et al. [6] provided an unsupervised baseline that uses hidden states to train binary classifiers for hallucination detection, and its open-source implementation offered a practical reference point for extension. Ogasa and Arase [7] focused on attention-layer features for hallucination span localization, while Binkowski et al. [8] applied spectral analysis to attention matrices, showing that eigenvalue patterns correlate with hallucination. Liu et al. [9] proposed a self-reflection mechanism leveraging attention-guided resampling for zero-shot detection, providing a lightweight approach without retraining or ground truth. Collectively, these studies illustrate the promise of internal signals but also highlight that evaluations have often been limited in scope and consistency.

Despite this progress, systematic comparisons across differ-

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ent types of internal representations remain scarce. Prior work typically focuses on a single signal—such as MLP activations or attention weights—and evaluates on narrow sets of models or tasks, leaving open questions about generalizability and the relative strengths of different representations.

To investigate this gap, we build upon the MIND dataset [6] but focus on developing a custom classifier architecture that systematically compares three representation paradigms: (1) features derived from MLP blocks, (2) features from attention blocks, and (3) a combination of both. Our framework is trained and evaluated across eight diverse LLMs, using metrics that include Area Under the ROC Curve (AUC), correlation with human annotations, and inference time.

Through this work, we aim to clarify which internal signals are most predictive of hallucination, advance interpretability in detection systems, and contribute toward standardized benchmarking of hallucination detection methods for trustworthy deployment of generative AI.

The remainder of this paper is organized as follows: Section II reviews related work on hallucination detection. Section III presents our research objectives and classifier design. Section IV outlines the dataset, experimental setup, and results. Section VI discusses limitations and future directions, and Section VII concludes with a summary of contributions.

II. Background and Related Work

Hallucination detection has evolved from early translation errors to a mature field encompassing LLMs, structured through three main trends: post-hoc verification, internal-state analysis, and spectral or self-reflective detection methods.

A. Early Hallucination Taxonomy

The concept gained initial academic notice with Lastras et al. [1], who defined hallucinations in neural machine translation. Ji et al. [10] and Rawte et al. [11] expanded this by categorizing hallucinations in LLM outputs — distinguishing types (e.g., intrinsic vs. extrinsic) and addressing evaluation gaps in benchmark datasets.

B. Post-Hoc Verification

Post-hoc methods validate generated text via retrieval, entailment, or knowledge graph checks. While effective, they often introduce latency, require external resources, and may fail in domains with sparse knowledge coverage [11].

C. Internal Representation-Based Methods

The INSIDE model demonstrated that encoder-decoder internal activations contain rich signals for hallucination detection, using covariance-based EigenScores and feature clipping [4]. Follow-up work from Ji et al. [12] showed LLM internals can predict hallucination risk before output, indicating a stronger self-awareness capability in models. Orgad et al. [5] used geometric structures in hidden states to detect factual errors, reinforcing the interpretability of internal features.

Su et al.'s MIND framework introduced an unsupervised binary classifier trained on hidden states — pioneering real-time detection without manual labels [6]. Ogasa and Arase extended this by localizing hallucinated spans, demonstrating interpretable attention features aligned with lexically hallucinated tokens [7].

D. Graph-Based and Self-Reflection Methods

Binkowski et al. proposed LapEigvals, which extracts Laplacian eigenvalues from attention maps, achieving state-of-the-art detection among attention-based methods [13]. Separately, Sriramanan et al. introduced LLM-Check, combining attention kernels, hidden activations, and token probabilities to create lightweight yet effective detectors [14]. These methods show that structural and interactive internal signals can be leveraged without heavy retraining or external knowledge.

E. Summary and Gap

While previous works such as MIND [6], INSIDE [4], and Eigen Attention [13] have advanced hallucination detection using internal features, they each present limitations. MIND shows inconsistent performance across different model sizes, particularly underperforming on architectures like LLaMA-13B. Eigen Attention introduces additional spectral complexity and may be sensitive to the structure of attention heads, while INSIDE requires carefully curated feature engineering. Moreover, many of these methods are optimized either for detection accuracy or model interpretability, but not both.

In this paper, we bridge that gap. We build a classifier and evaluate three different architectures. Our experiments show that a lightweight classifier trained on averaged hidden representations, regularized with dropout and batch normalization, offers both robustness and generalizability. The proposed method achieves stable performance across model scales (7B to 70B) and demonstrates stronger alignment with human-annotated hallucination labels compared to prior baselines. These results support our approach as a more effective and scalable alternative for hallucination detection in LLMs.

III. Research Objectives and Methodology

A. Research Objectives

This study aims to advance hallucination detection in LLMs through the development of an efficient and scalable classifier that exploits internal model signals. Our primary goals are:

- To investigate which internal representation — Attention-derived features, MLP-derived features, or a combination — are most predictive of hallucinations.
- To evaluate the performance of these detection paradigms across a diverse set of LLMs varying in size and architecture.
- To develop a standalone classifier that maintains robustness and generalizability while remaining lightweight enough for real-time deployment.

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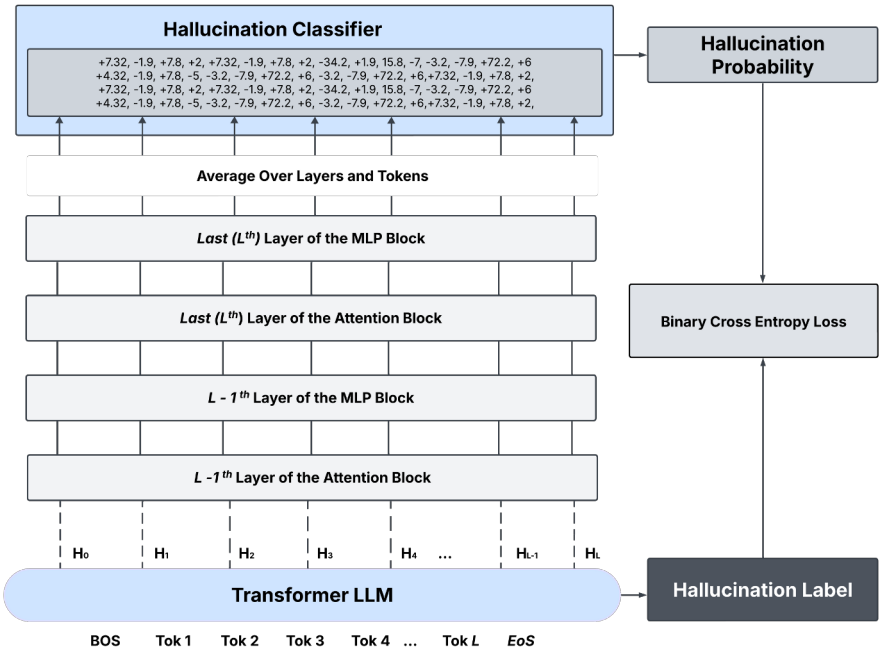


Fig. 1: Overall architecture of the hallucination classification pipeline. Token-level representations are extracted from the last two layers of the LLM's MLP and/or Attention blocks, aggregated, and passed to a binary classifier.

B. Transformer Architecture Foundation

Modern LLMs are built upon the Transformer architecture, which comprises alternating blocks of multi-head self-attention and feed-forward MLP layers. During autoregressive generation, each input prompt undergoes a series of transformations through these layers, resulting in hierarchical hidden representations. The attention mechanism enables contextual alignment by assigning dynamic weights to tokens, while the MLP blocks enrich token-wise representations through nonlinear projections.

At layer ℓ, the hidden representation $\mathbf{H}^{(\ell)} \in \mathbb{R}^{\mathbf{m}\times\mathbf{d}}$ is updated as:

$$\mathbf{H}' = \text{LayerNorm}(\mathbf{H}^{(\ell-1)} + \mathbf{A}^{(\ell)}(\mathbf{H}^{(\ell-1)})) \quad (1)$$

$$\mathbf{H}^{(\ell)} = \text{LayerNorm}(\mathbf{H}' + \mathbf{M}^{(\ell)}(\mathbf{H}')) \quad (2)$$

where $\mathbf{A}^{(\ell)}(\cdot)$ and $\mathbf{M}^{(\ell)}(\cdot)$ denote the attention and MLP blocks at layer ℓ, respectively, m is the number of tokens, and d is the hidden dimension.

C. Embedding Extraction and Classifier Variants

To construct features for hallucination detection, we extract token-level hidden states from the final two transformer layers (L and $L-1$). These are averaged across tokens to form a compact representation:

$$\mathbf{z} = \frac{1}{2m}\sum_{i=1}^{m}\left(\mathbf{H}_i^{(L-1)} + \mathbf{H}_i^{(L)}\right) \quad (3)$$

where $\mathbf{H}_i^{(L)}$ represents the vector of elements in i-th block.

Although we employ mean pooling for simplicity, our design allows for weighted token aggregation:

$$\mathbf{z} = \sum_{i=1}^{m}\alpha_i\mathbf{H}_i^{(L-1)} + \beta_i\mathbf{H}_i^{(L)}, \quad \sum_{i=1}^{m}\alpha_i + \beta_i = 1 \quad (4)$$

Here, α_i and β_i are distribution weights, either uniform or learnable. Future extensions can explore adaptive pooling strategies.

We define three types of features:

- **MLP-only**: $\mathbf{z}_{\text{MLP}}$ from the MLP block outputs.
- **Attention-only**: $\mathbf{z}_{\text{Attn}}$ from the attention block outputs.
- **Hybrid**: $\mathbf{z}_{\text{Hybrid}} = [\mathbf{z}_{\text{MLP}}; \mathbf{z}_{\text{Attn}}]$ via concatenation.

This design enables comparative analysis while facilitating late fusion of orthogonal representations. The hybrid approach leverages both inter-token dependencies (attention) and non-linear feature refinements (MLP), leading to stronger detection performance (as shown in Sec. IV).

Each embedding $\mathbf{z} \in \mathbb{R}^d$ or $\mathbb{R}^{2d}$ is passed into a classifier $f_\theta(\mathbf{z})$ producing binary logits. The hallucination probability is:

$$\hat{y} = \text{softmax}(f_\theta(\mathbf{z})) \quad (5)$$

We optimize binary cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)] \quad (6)$$

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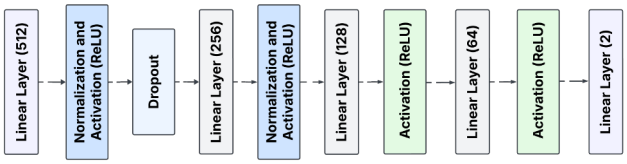


Fig. 2: Classifier architecture used for hallucination detection. It is a multi-layer perceptron with ReLU activations, normalization, and dropout, progressively reducing dimensionality.

D. Classifier Design

To capture nuanced patterns from the embeddings, we employ a five-layer feed-forward neural network (MLP), shown in Fig. 2. The classifier includes linear transformations, layer normalization, dropout, and ReLU activations. This structure balances complexity and generalization.

The architecture is recursively defined as:

$$\begin{aligned} f_1(\mathbf{z}) &= \phi(\mathbf{w}_1\mathbf{z} + \mathbf{b}_1) \\ f_k(\mathbf{z}) &= \phi(\mathbf{w}_k f_{k-1}(\mathbf{z}) + \mathbf{b}_k), \quad k = 2 \cdots 5 \\ f(\mathbf{z}) &= \mathbf{w}_5 f_4(\mathbf{z}) + \mathbf{b}_5 \end{aligned} \tag{7}$$

where $\phi(\cdot)$ denotes ReLU activation, and all $\mathbf{w}_k$, $\mathbf{b_k}$, $k = 1, 2, \cdots 5$, are trainable parameters.

E. Classifier Pipeline Overview

The complete detection pipeline is visualized in Fig. 1. Token-level representations from the final two transformer layers are aggregated into sentence-level embeddings, which are passed into a dedicated classifier. The classifier outputs a hallucination probability, trained using supervised labels and evaluated using AUC and Pearson correlation.

F. Efficiency and Deployment

A central design goal is model-agnostic deployment. Our pipeline avoids modifying or fine-tuning the LLM. Instead, it operates on top of extracted hidden states. This allows reuse of the classifier across models such as Falcon, OPT, LLaMA, and GPT-J. Since we only extract the final two layers, runtime overhead is minimal. The classifier's shallow depth ensures rapid inference, making it suitable for real-time or edge settings. Moreover, because the hallucination prediction is disentangled from the decoder head, the system is robust to model-specific generation dynamics.

IV. Dataset and Experimental Results

A. Dataset

We evaluate our hallucination detection classifier using the MIND dataset [15], which is derived from the HELM benchmark and includes outputs from six diverse LLMs: Falcon-40B, GPT-J-6B, LLaMA-based models (LLB-7B, LLC-7B, LLC-13B), and OPT-7B. The dataset comprises 3,582 total sentences, each labeled with binary hallucination annotations based on factual consistency with the source Wikipedia passage. We use the HuggingFace Transformers library[1]. to download and load the models for feature extraction and evaluation. The dataset and codebase are publicly available on GitHub[2].

TABLE I: Statistics of the MIND Dataset across Multiple LLMs

Model	#Sentence	#H=1	#H=0	#Passage
Falcon-40B	521	261	260	196
GPT-J-6B	572	172	400	208
LLB-7B	565	243	322	198
LLC-7B	617	308	309	202
LLC-13B	596	329	267	207
OPT-7B	566	181	385	213
LLB-70B	565	243	322	198
LLC-70B	617	308	309	202
Total	**3582**	**1494**	**2088**	**1224**

Human annotators assess each generated sentence, focusing only on factual inaccuracies and disregarding grammatical issues or stylistic preferences. Annotations are recorded as `H=1` for hallucinated and `H=0` for non-hallucinated examples. The dataset provides a balanced distribution of examples, allowing our classifier to learn subtle discriminative features.

B. Experimental Setup

We evaluate our hallucination classifier using extracted hidden state embeddings from eight LLMs: Falcon-40B, GPT-J-6B, OPT-7B, and multiple LLaMA variants. For each model, we use the HuggingFace Transformers library to obtain token-wise hidden states, averaging across the last two layers to form a fixed-size sentence embedding. The embeddings are then used to train a feed-forward neural network classifier.

We investigate three different architectures for hallucination detection:

- **LIRa, Attention-only:** Embeddings derived from attention block outputs.
- **LIRp, MLP-only:** Embeddings derived from MLP (feed-forward) layers.
- **LIRx, Hybrid:** A combination of both attention and MLP-derived representations.

The classification model is a multilayer perceptron with five hidden layers and ReLU or GELU activation. Each layer is followed by batch normalization and dropout to improve generalization.

Hyperparameters: Table II summarizes the training configuration used across all classifier runs.

The model is trained and evaluated on an NVIDIA GPU with float16 precision and 8-bit quantized LLM inference for memory efficiency. For each LLM, we extract sentence-level and passage-level performance metrics, including AUC and Pearson correlation with human annotations.

[1]https://huggingface.co/meta-llama/Llama-2-7b
[2]https://github.com/oneal2000/MIND

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TABLE II: Hyperparameter Configuration for Classifier Training

Parameter	Value
Optimizer	AdamW
Learning Rate	5×10^{-4}
Weight Decay	1×10^{-5}
Batch Size	2
Epochs	200
Early Stopping Patience	50
Dropout Rate	0.3
Scheduler	CosineAnnealingLR
Precision	Float16
Quantization	8-bit (BitsAndBytes)

C. Performance Evaluation and Baselines

To assess the effectiveness of our hallucination detection classifier, we employ two main evaluation metrics: Area Under the Receiver Operating Characteristic Curve (AUC) and Pearson correlation. The AUC captures the classifier's ability to distinguish hallucinated from non-hallucinated outputs across various thresholds, while the Pearson correlation measures the strength of linear association between the predicted probabilities and human annotations. We report these metrics at both sentence-level and passage-level granularity.

In addition to our proposed classifier, we compare performance against several state-of-the-art reference-free hallucination detection baselines:

- **Predictive Probability (PP)** [16]: This technique analyzes the probability scores of each token generated by an LLM. Various aggregation strategies are used to summarize these scores into a sentence-level feature, including maximum (PP_{max}), minimum (PP_{min}), and mean (length-normalized PP).
- **Predictive Entropy (PE)** [17]: This metric quantifies uncertainty in the model's token-level distribution using entropy. For each token t_i, entropy is computed as:

$$\text{PE}(t_i) = -\sum_{\tilde{w} \in \mathcal{V}} p_i(\tilde{w}) \log p_i(\tilde{w}), \tag{8}$$

  where $\mathcal{V}$ is the vocabulary and $p_i(\tilde{w})$ is the probability assigned to token $\tilde{w}$. Sentence-level scores are derived using PE_{max}, PE_{min}, and mean length-normalized PE.
- **SelfCheckGPT (SCG)** [16]: This method assesses the factual consistency of multiple generations from the same prompt. If the generations are internally consistent, the model is less likely to hallucinate. It includes variants based on BERTScore, QA response similarity, natural language inference (NLI), and n-gram overlap.
- **SAPLMA** [18]: This method trains a dedicated hallucination classifier using internal activation values (hidden states) extracted from LLMs, allowing for supervised detection without requiring reference texts.
- **EUBHD** [19]: Enhanced Uncertainty-Based Hallucination Detection leverages multiple uncertainty-aware features from the model's output distribution to flag hallucinated generations.
- **GPT-4 HDM** [20]: This baseline uses GPT-4 as an external verifier. It prompts GPT-4 to evaluate whether outputs from other LLMs are hallucinated using a structured instruction template.
- **MIND** [6]: This baseline uses a multi-granularity alignment framework that detects hallucinations by modeling interactions between the generated answer and its supporting context. It combines sentence-level and token-level alignment signals using a dual-encoder architecture with cross-attention mechanisms.

D. LIRp: MLP-Only Embeddings

Table III and Table IV present the AUC and Pearson correlation scores, respectively, for hallucination detection using embeddings derived solely from the MLP (feed-forward) layers of the model. This approach isolates the contribution of non-attentional components, allowing an assessment of how much hallucination-related signal is retained in the feed-forward sublayers. The classifier achieves moderate performance, with AUC and correlation scores demonstrating consistent but limited alignment with ground truth annotations. Notably, performance at the passage level tends to be slightly higher than at the sentence level, suggesting that the MLP representations may encode broader contextual consistency cues more effectively than fine-grained local hallucination signals.

E. LIRa: Attention-Only Embeddings

Results for the attention-only approach, shown in Table V and Table VI, reveal the performance when using representations extracted from the attention blocks alone. Compared to LIRp, LIRa generally yields improved AUC and correlation metrics, particularly at the sentence level. This suggests that attention-based embeddings capture more direct alignment between generated content and source evidence, benefiting fine-grained hallucination detection. The correlation trends also highlight the attention layers' stronger sensitivity to localized factual inconsistencies.

F. LIRx: Hybrid Embeddings (Attention + MLP)

Tables VII and VIII report results for the hybrid approach, which combines attention- and MLP-derived embeddings. This fusion enables the classifier to leverage both fine-grained alignment from attention layers and broader contextual cues from feed-forward layers. As expected, LIRx achieves the best overall performance across all models, with notable improvements in both AUC and correlation scores at both sentence and passage levels. The hybrid approach demonstrates that integrating complementary features from attention and feed-forward layers leads to more robust and consistent hallucination detection, especially in complex outputs where hallucinations manifest in both localized and distributed

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TABLE III: Performance of LIRp Classifier at Sentence and Passage Levels AUC

	LIRp Classifier Sentence Level AUC						**MIND Classifier Passage Level AUC**					
	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
MIND	0.790	0.877	0.788	0.604	0.676	0.884	0.889	0.960	0.905	0.718	0.855	0.945
LIRp (Ours)	0.810	0.804	0.834	0.832	0.846	0.845	0.810	0.820	0.835	0.856	0.856	0.846

TABLE IV: LIRp Sentence Level and Passage Level Correlation

	LIRp Sentence Level Correlation						**LIRp Passage Level Correlation**					
Baselines	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
MIND	0.503	0.524	0.486	0.427	0.494	0.476	0.525	0.530	0.491	0.382	0.478	0.564
LIRp (Our)	0.536	0.529	0.579	0.582	0.600	0.609	0.497	0.510	0.555	0.590	0.574	0.584

TABLE V: Performance of LIRa Classifier at Sentence and Passage Levels AUC

	LIRa Classifier Sentence Level AUC						**LIRa Classifier Passage Level AUC**					
Baselines	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
MIND	0.790	0.877	0.788	0.604	0.676	0.884	0.889	0.960	0.905	0.718	0.855	0.945
LIRa (Our)	0.814	0.888	0.845	0.839	0.832	0.841	0.815	0.814	0.850	0.867	0.865	0.855

TABLE VI: Performance of LIRa Classifier at Sentence and Passage Levels Correlation

	LIRa Classifier at Sentence Level Correlation						**LIRa Passage Level Correlation**					
Baselines	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
MIND	0.503	0.524	0.486	0.427	0.494	0.476	0.525	0.530	0.491	0.382	0.478	0.564
LIRa (Our)	0.814	0.788	0.845	0.839	0.832	0.841	0.815	0.814	0.850	0.867	0.845	0.855

TABLE VII: Performance of LIRx Classifier at Sentence and Passage Levels AUC

	LIRx Classifier Sentence Level AUC						**LIRx Classifier Passage Level AUC**					
Baselines	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
PE-max	0.648	0.750	0.685	0.444	0.493	0.726	0.835	0.888	0.840	0.593	0.699	0.885
PE-min	0.576	0.704	0.588	0.316	0.441	0.723	0.712	0.760	0.759	0.592	0.641	0.808
PP-max	0.575	0.707	0.587	0.301	0.417	0.730	0.706	0.759	0.754	0.534	0.661	0.810
PP-min	0.555	0.741	0.603	0.373	0.448	0.712	0.765	0.853	0.797	0.530	0.687	0.838
LNPP	0.533	0.693	0.610	0.267	0.354	0.721	0.701	0.804	0.776	0.552	0.595	0.837
LNPE	0.544	0.698	0.618	0.335	0.411	0.702	0.715	0.811	0.778	0.588	0.670	0.835
SCG-MQAG	0.541	0.787	0.640	0.404	0.461	0.759	0.728	0.883	0.820	0.667	0.728	0.859
SCG-NG	0.522	0.755	0.537	0.265	0.311	0.749	0.712	0.858	0.716	0.498	0.577	0.834
SCG-BS	0.642	0.742	0.618	0.303	0.376	0.659	0.743	0.817	0.763	0.476	0.594	0.760
SCG-NLI	0.685	0.868	0.764	0.583	0.657	0.810	0.812	0.938	0.890	0.756	0.795	0.910
SAPLMA	0.513	0.699	0.578	0.305	0.407	0.621	0.724	0.829	0.782	0.518	0.627	0.748
EUBHD	0.751	0.759	0.648	0.466	0.481	0.756	0.866	0.856	0.786	0.669	0.713	0.855
GPT4-HDM	0.633	0.784	0.658	0.411	0.513	0.797	0.815	0.918	0.863	0.590	0.777	0.920
MIND	0.790	0.877	0.788	0.604	0.676	0.884	0.889	0.960	0.905	0.718	0.855	0.945
LIRx (Our)	0.937	0.939	0.923	0.891	0.904	0.917	0.943	0.968	0.930	0.914	0.914	0.959

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TABLE VIII: Performance of LIRx Classifier at Sentence and Passage Levels Correlation

Baselines	LIRx Classifier at Sentence Level Correlation						LIRx Passage Level Correlation					
	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
PE-max	0.241	0.084	0.203	0.238	0.203	0.057	0.311	0.266	0.256	0.226	0.196	0.218
PE-min	0.120	−0.032	0.015	−0.017	0.044	0.060	0.086	−0.065	0.093	0.238	0.083	0.082
PP-max	0.119	−0.047	0.015	−0.040	−0.034	0.073	0.084	−0.067	0.084	−0.043	0.143	0.094
PP-min	0.150	0.206	0.184	0.109	0.129	0.181	0.182	0.245	0.223	0.094	0.167	0.119
LNPP	0.067	0.077	0.175	−0.097	−0.034	0.208	0.071	0.160	0.230	0.219	0.117	0.242
LNPE	0.094	0.038	0.146	0.083	0.086	0.122	0.102	0.126	0.202	0.287	0.226	0.188
SCG-MQAG	0.037	0.231	0.182	0.182	0.186	0.215	−0.085	0.315	0.228	0.290	0.258	0.299
SCG-NG	0.100	0.177	0.059	−0.048	−0.102	0.117	0.058	0.222	0.079	−0.107	−0.135	0.102
SCG-BS	0.229	0.075	0.127	−0.014	−0.038	−0.056	0.247	0.129	0.145	−0.014	−0.050	−0.073
SCG-NLI	0.379	0.409	0.381	0.309	0.384	0.331	0.406	0.422	0.439	0.415	0.401	0.409
SAPLMA	0.021	0.046	−0.002	0.000	0.013	−0.141	0.090	0.130	0.027	−0.071	0.052	−0.143
EUBHD	0.316	0.106	0.117	0.212	0.104	0.142	0.345	0.155	0.169	0.308	0.051	0.134
GPT4-HDM	0.135	0.110	0.009	0.077	0.120	0.168	0.081	0.254	0.188	0.053	0.188	0.237
MIND	0.503	0.524	0.486	0.427	0.494	0.476	0.525	0.530	0.491	0.382	0.478	0.564
LIRx (Our)	0.829	0.829	0.818	0.839	0.789	0.808	0.706	0.707	0.699	0.867	0.672	0.693

forms. Furthermore, as shown in Table IX, LIRx consistently outperforms the individual LIRp and LIRa approaches across all evaluated LLMs, confirming that the combination of both representation types is more effective than relying on either one alone.

TABLE IX: AUC comparison of our three different classifiers

Approach	Falcon	GPT-J	LLB-7B	LLC-13B	LLC-7B	OPT-7B
LIRp	0.810	0.804	0.834	0.832	0.846	0.845
LIRa	0.814	0.888	0.845	0.839	0.832	0.841
LIRx	0.937	0.939	0.923	0.891	0.904	0.917

G. Latency Analysis

Table XI reports the average inference time for each evaluated model when used with our hallucination classifier. Here, inference time refers to the total duration required to process an input through both the LLM (to extract the final two layers' hidden states) and the classifier to produce a hallucination prediction.

OPT-7B records the fastest inference time (0.0362 s), with LLC-7B and LLB-7B close behind. The 70B-parameter LLaMA-2 models are the slowest (over 0.11 s), while Falcon and GPT-J show moderate latency.

These results illustrate the trade-off between model size and inference efficiency. While the classifier adds minimal overhead, the underlying LLM significantly impacts latency, which is an important consideration for real-time or resource-constrained deployments.

V. Ablation Study

To evaluate the individual contributions of attention-based and MLP-based features, we conducted ablation experiments using two variants of our hybrid LIRx model: LIRa (Attention-only) and LIRp (MLP-only). These variants isolate the impact of each component by removing either the MLP or attention-derived features prior to classification.

As shown in Tables VI–VIII, both LIRa and LIRp underperform the full LIRx classifier across nearly all large language models and evaluation metrics. Notably, LIRp suffers in sentence-level AUC scores for smaller models like GPT-J and LLC-7B, likely due to its inability to model token interactions directly. In contrast, LIRa struggles with capturing higher-order semantic abstractions, which becomes evident in its lower passage-level correlation scores on models like Falcon

TABLE X: AUC and Correlation comparison of our classifiers on 70B parameter models

Classifier at Sentence Level AUC			Classifier at Passage Level AUC		
Approach	LLB-70B	LLC-70B	Approach	LLB-70B	LLC-70B
LIRp	0.834	0.864	LIRp	0.839	0.867
LIRa	0.826	0.868	LIRa	0.810	0.887
LIRx	0.900	0.937	LIRx	0.803	0.907

Classifier at Sentence Level Correlation			Classifier at Passage Level Correlation		
Approach	LLB-70B	LLC-70B	Approach	LLB-70B	LLC-70B
LIRp	0.588	0.634	LIRp	0.568	0.615
LIRa	0.567	0.642	LIRa	0.499	0.643
LIRx	0.763	0.812	LIRx	0.633	0.670

TABLE XI: Average Inference Time (in seconds) for Each Model with the Classifier

Model	Time (s)	Model	Time (s)
Falcon	0.0580	LLC-7B	0.0438
GPT-J	0.0735	OPT-7B	0.0362
LLB-7B	0.0437	LLB-70B	0.1139
LLC-13B	0.0571	LLC-70B	0.1112

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and LLaMA-2.

These findings underscore the complementary nature of attention and MLP blocks in transformer architectures. The attention mechanism is adept at modeling local and global alignments between generated text and source context, while the MLP layers enrich this signal through nonlinear transformations that aid in detecting semantic drift. Their combination in LIRx results in more robust and generalizable representations for hallucination classification.

While either component can serve as a weak learner on its own, their integration consistently boosts performance, confirming that the hybrid representation is essential for capturing both structural and semantic inconsistencies. This highlights the value of multi-granular feature aggregation for improving alignment-aware hallucination detection.

VI. Limitations and Future Work

Despite the strong performance of the proposed LIRx classifier, certain limitations remain. The evaluation is restricted to binary hallucination detection, limiting granularity for real-world interpretability. Furthermore, while the classifier is lightweight, overall inference time is still constrained by the underlying LLM, which can challenge deployment in latency-sensitive environments.

Future work will explore adaptive multi-layer fusion techniques to leverage richer hidden state representations, extend the classifier to support fine-grained hallucination categorization, and investigate model compression or distillation strategies to reduce runtime without compromising accuracy.

VII. Conclusion

This paper introduces a hallucination detection framework that leverages embeddings extracted from both attention and MLP layers of large language models. We evaluate three architectural variants — LIRa (attention-only), LIRp (MLP-only), and LIRx (hybrid) — across eight prominent LLMs using both sentence-level and passage-level detection strategies.

Our experimental results show that the hybrid LIRx classifier consistently outperforms not only the individual LIRa and LIRp variants but also all baseline methods, including MIND, SelfCheckGPT, and uncertainty-based detectors. LIRx achieves the highest AUC and Pearson correlation scores across all evaluated models, confirming the complementary value of attention-derived alignment signals and MLP-based semantic abstraction. Theoretical insights and ablation studies further reinforce the effectiveness of combining these representations. In addition, latency analysis demonstrates that our classifier adds minimal overhead to inference, underscoring its practicality for real-time deployment.

Together, these findings highlight that dual-source embeddings provide a more robust and generalizable basis for hallucination detection than existing baselines. Our work establishes a new performance benchmark and lays the groundwork for future efforts to design faster, scalable, and semantically rich classifiers that ensure factual fidelity in language model outputs.

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